



Perceptron Learning Algorithm

Algorithm: Perceptron Learning Algorithm

$P \leftarrow$ inputs with label 1;

$N \leftarrow$ inputs with label 0;

Initialize $\mathbf{w}$ randomly;

while !convergence **do**

 Pick random $\mathbf{x} \in P \cup N$;

if $\mathbf{x} \in P$ and $\mathbf{w} \cdot \mathbf{x} < 0$ **then**

$\mathbf{w} = \mathbf{w} + \mathbf{x}$;

end

if $\mathbf{x} \in N$ and $\mathbf{w} \cdot \mathbf{x} \geq 0$ **then**

$\mathbf{w} = \mathbf{w} - \mathbf{x}$;

end

end

//the algorithm converges when all the
inputs are classified correctly



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Step 1: Divide the Training Data

- **P** = Set of positive examples (label = 1)
- **N** = Set of negative examples (label = 0)

For the OR gate:

Input	Label
(0,0)	0
(0,1)	1
(1,0)	1
(1,1)	1

So,

$$P = \{(0, 1), (1, 0), (1, 1)\}$$

$$N = \{(0, 0)\}$$



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Step 2: Initialize Weights Randomly

Choose random values for the weights:

$$w = [w_0, w_1, w_2]$$

Example:

$$w = [0.2, -0.5, 0.7]$$

Initially, these weights may classify samples incorrectly.

Step 4: Select a Random Training Sample

Pick any training example:

$$x \in P \cup N$$

Example:

$$x = (1, 0)$$



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Step 5: Check Positive Examples

If the selected sample belongs to the positive class:

$$x \in P$$

but the perceptron predicts:

$$w \cdot x < 0$$

then it is wrongly classified as negative.

Update:

$$w = w + x$$



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Step 6: Check Negative Examples

If the selected sample belongs to the negative class:

$$x \in N$$

but the perceptron predicts:

$$w \cdot x \geq 0$$

then it is wrongly classified as positive.

Update:

$$w = w - x$$



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$$\cos\alpha = \frac{\mathbf{w}^T \mathbf{x}}{\|\mathbf{w}\| \|\mathbf{x}\|} \quad \left| \quad \cos\alpha \propto \mathbf{w}^T \mathbf{x} \right.$$

So if $\mathbf{w}^T \mathbf{x} > 0 \Rightarrow \cos\alpha > 0 \Rightarrow \alpha < 90$

Similarly, if $\mathbf{w}^T \mathbf{x} < 0 \Rightarrow \cos\alpha < 0 \Rightarrow \alpha > 90$

